

PHYO THANT KYAW

Chiang Rai, Thailand | +66855129146 | phyothantkyaw22.pku@gmail.com | github.com/pt22-mfu | ptportfolio-aw35t6wq.manus.space

Nationality: Myanmar | Visa: Non-Immigrant ED Plus | **Authorized: 16-week Co-op, No Sponsorship Required**

Eligible: 1-Year Post-Grad Extension + BOI IT Specialist Tier | Languages: English (Professional), Thai (Basic), Burmese (Native)

SUMMARY

Python backend developer and GenAI engineer specializing in RAG pipelines, LangChain orchestration, and LLM integration—deploying production AI systems via Streamlit and FastAPI.

Seeking a 16-week Co-op (January 2027) to embed AI models into scalable, production-grade web platforms at a Tier-1 Thai tech organization.

EDUCATION

Mae Fah Luang University (MFU) | B.Eng., Computer Engineering | GPAX: **3.84**

Expected 2027

Mandatory 16-week Cooperative Education: **January 2027**

Mandalay Technological University (MTU) | B.Eng., Electronics Engineering | GPA: 3.04 *Dec 2016 – Sep 2019 (Year 3)*

TECHNICAL PROJECTS

Chiang Mai Tri-Node Zone: Geo-Predictive AI & Mitigation Platform | [Live Demo](#) | [GitHub](#)

Mar – May 2026

- **Architected** an end-to-end geospatial AI web platform in Python integrating multi-source environmental datasets into a unified Streamlit dashboard.
- **Engineered** an LLM-powered Mitigation Copilot using LangChain + Google Gemini, generating real-time actionable crisis-management strategies from live situational inputs.
- **Embedded** an interactive geospatial threat map within Streamlit, rendering 3-node Z-score overlays with drill-down zone analysis.
- **Deployed** the full AI platform live to Streamlit Cloud, integrating geospatial inference, LLM reasoning, and interactive UI in a single-application architecture.

MFU PM2.5 GeoAI Warning System — Senior Project | [Live Demo](#) | [GitHub](#)

Jan – Jul 2026

- **Engineered** automated ETL pipelines integrating OpenWeatherMap and NASA FIRMS APIs for scheduled, normalized multi-source environmental data ingestion.
- **Benchmarked** 4 candidate models against a Fire-Integrated feature set, with LightGBM emerging as champion at **85.90% R²** (MAE: 3.21) by feature-engineering wildfire activity variables with meteorological predictors.
- **Deployed** a live, interactive 5-day PM2.5 forecast dashboard on Streamlit Cloud serving real-time air quality predictions to MFU campus stakeholders.
- **Implemented** model-monitoring hooks and automated data refresh routines ensuring forecast outputs remain current with zero manual intervention post-deployment.

EXPERIENCE

Work Assistant — Web & AI Projects | MFU Learning Innovation Institute (MLii)

May 2026 – Jul 2026

- **Architected** a Zero-Budget Hybrid AI Evaluation Engine for a clinical web platform, combining a fast rule-based matcher (< 500ms) with Google Gemini 1.5 Flash LLM for semantic assessment.
- **Authored** the comprehensive Software Requirements Specification (SRS) and developed a functional UI prototype using Next.js 14 and Tailwind CSS.
- **Designed** a Thai PDPA-compliant secure data flow utilizing Supabase (PostgreSQL) and Upstash Redis caching to enforce strict API rate limits.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript

AI & ML: LangChain, Google Gemini API, RAG Pipelines, PyTorch, TensorFlow, LightGBM, XGBoost, Scikit-learn

Web & Backend: Streamlit, FastAPI, RESTful API Design, Web Integration

Cloud & Tools: Google Cloud Platform (GCP), Git, GitHub

Certifications: IBM Data Engineering Professional | Google Data Analytics | Google Cloud: Data Lakes & Warehouses | UC Davis: SQL for Data Science | Google AI Essentials, Prompting Essentials